10 Minute Talk: RNO-G Hpols

8/6/25

Outline

01 Important Vocab

Definitions!

02 Hpol Introduction

Why do we need hpols?
What's their current design?

03 Plots

Current hpol performance

04 Calibration Efforts

How have antennas been calibrated and what other calibration strategies might we consider?

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01

Important Vocab

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Vocabulary

Impedance

A measure of opposition of a current at a particular frequency.
Analogous to friction.

Reflection Coefficient (S11/Γ)

Accounts for the impedance mismatch between transmission line and antenna

Function of Z (impedance):

$$\Gamma = \frac{Z_{in} - Z_o}{Z_{in} + Z_o}$$

Voltage Standing Wave Ratio (VSWR)

Ratio of the peak voltage to the minimum voltage.

Function of Γ (S11):

$$\mathrm{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

Gain

Directionally and frequency dependent, unitless ratio that measures the signal received by an antenna, relative to an isotropic radiator

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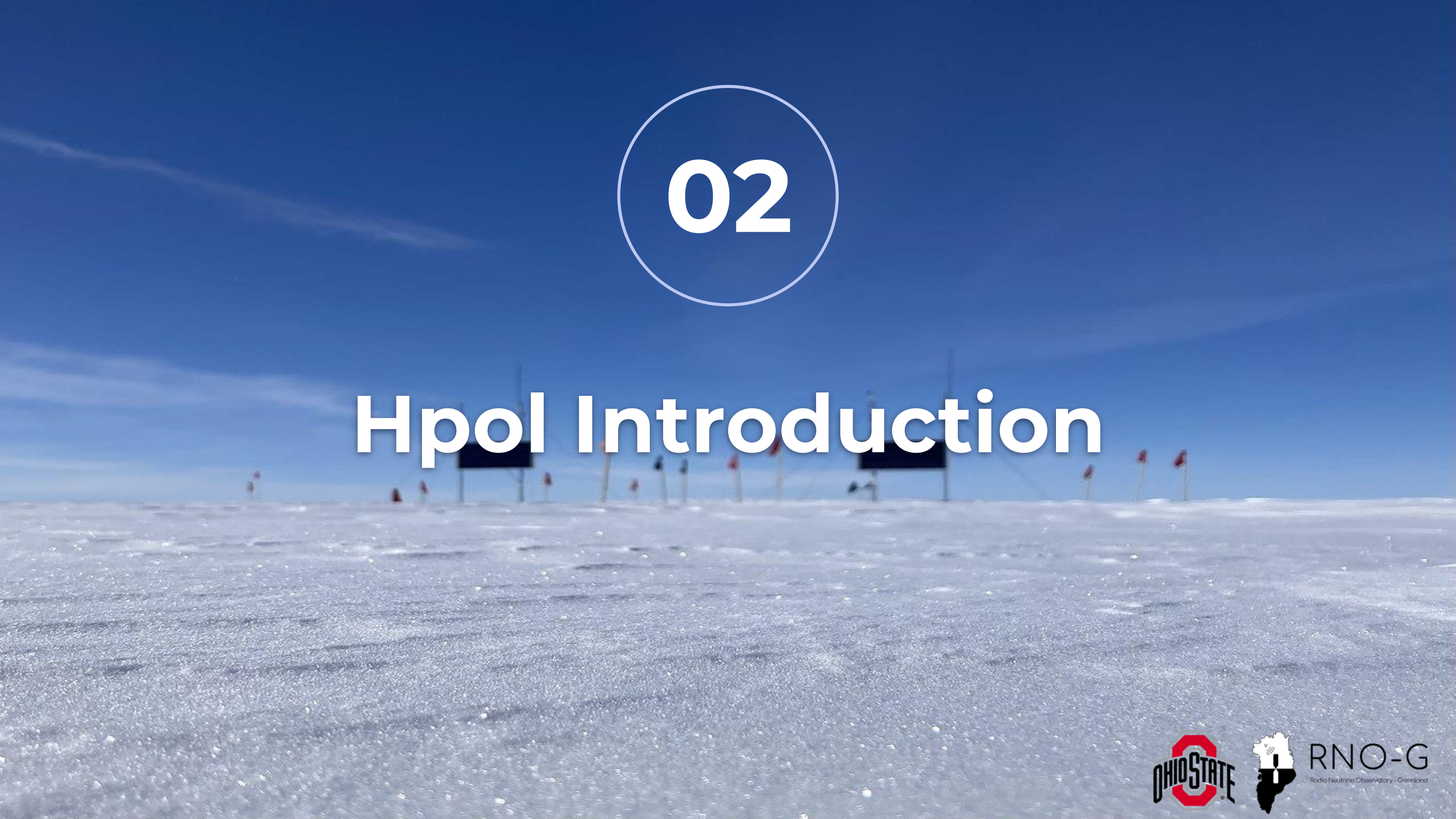

02

Hpol Introduction

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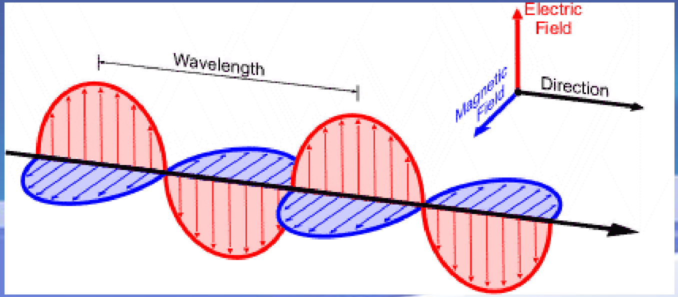
What are hpol antennas?

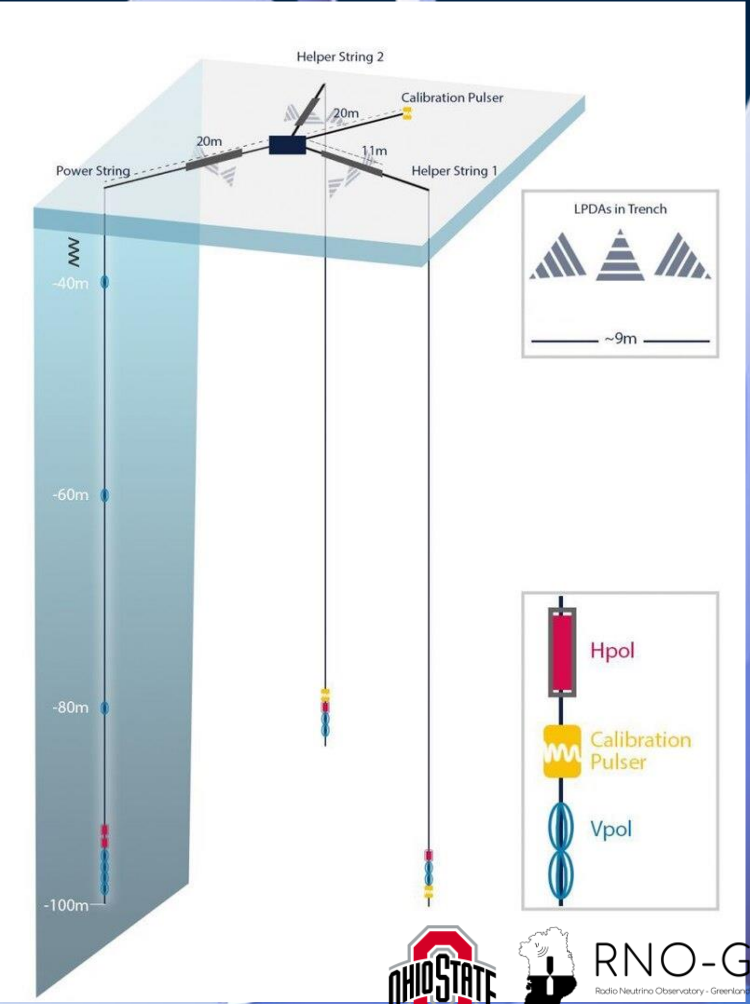
These are horizontally polarized antennas:

- Found on each string in borehole
- Difficult to design a horizontal antenna for a vertical slot
- Slot antenna design: interchanges electric and magnetic fields

Why do we need hpol antennas?

- Crucial for reconstruction of neutrino arrival direction
- Need both vpol and hpol





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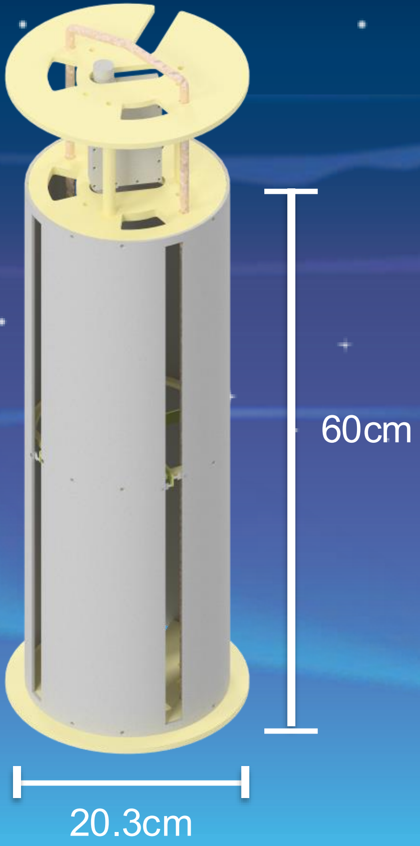
Older Hpol Design – Deployed for 2024 season

Technical Qualities

- Azimuthally-symmetric in gain
- (simulated) 3:1 VSWR in-ice bandwidth of 300-850 MHz
- 50Ω port



Physical Properties

- 60cm aluminum cylinder sheet
- Quad slots of same length
- Outer diameter 20.3cm
- Weighs 3.95kg
- Nylon structural end caps

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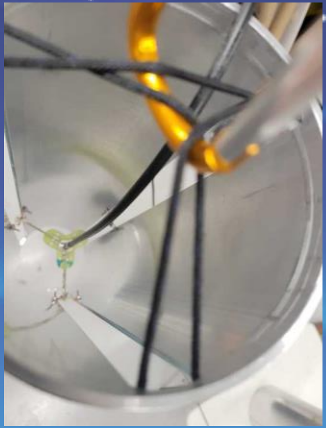
New Hpol Design – Work in progress

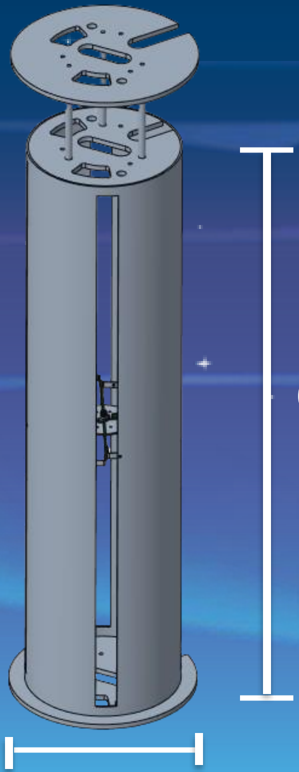
Technical Qualities

- Azimuthally-symmetric in gain
- VSWR bandwidth will change (TBD)
- Port may change (TBD)

Everett's WIP CAD model

Photo Creds: Penn State Group





Physical Properties

- 60cm aluminum cylinder sheet
- Tri or quad (TBD) slots of same length
- Outer diameter 12.7cm
- Weighs less
- Nylon structural end caps

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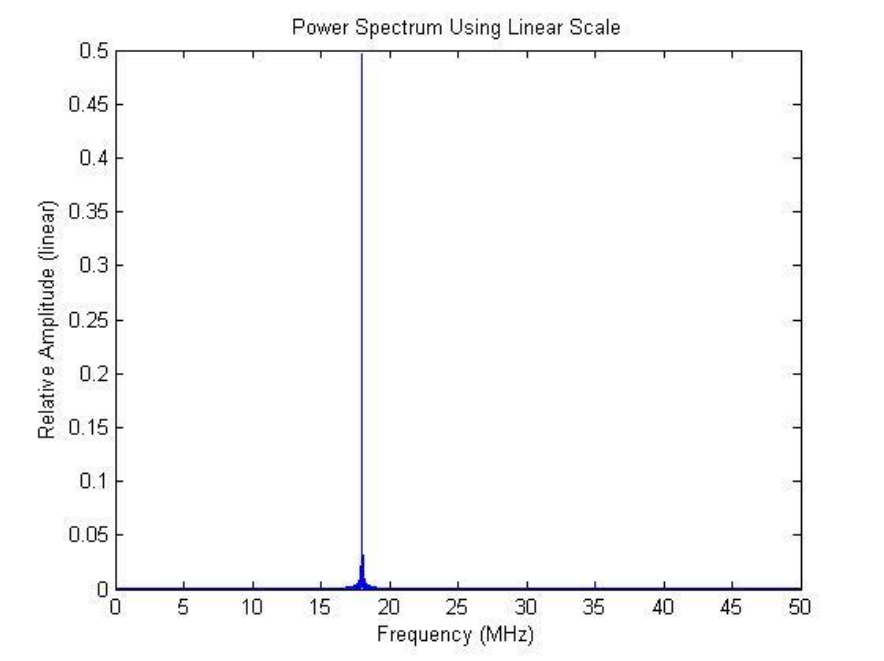
03

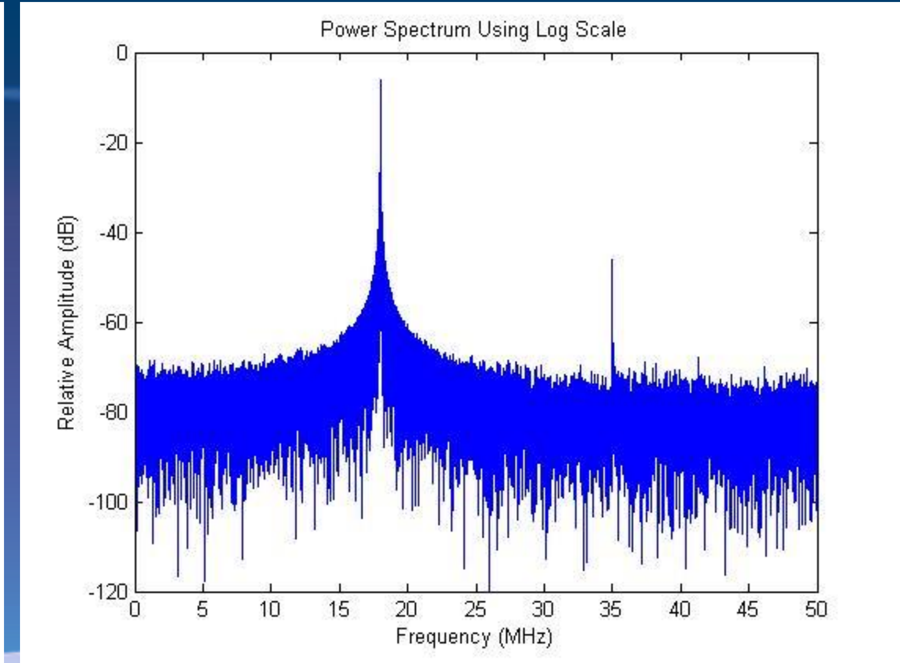
Plots

IMPORTANT: These are all for the 2020-2021 hpol design

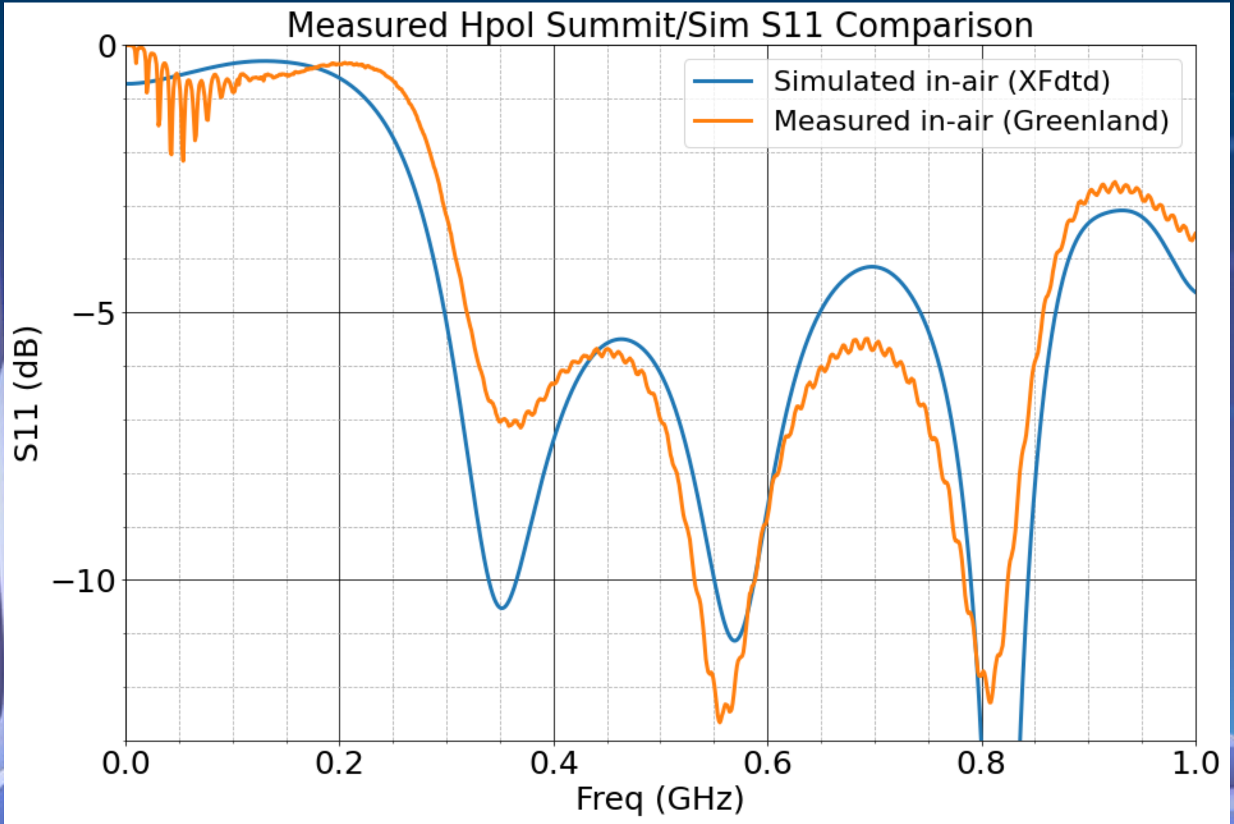
dB Scale

Why dB scale is beneficial:





S11 vs Frequency (in-air)



- S11 of -∞ dB -> no reflection
- S11 of 0 db -> 100% reflection

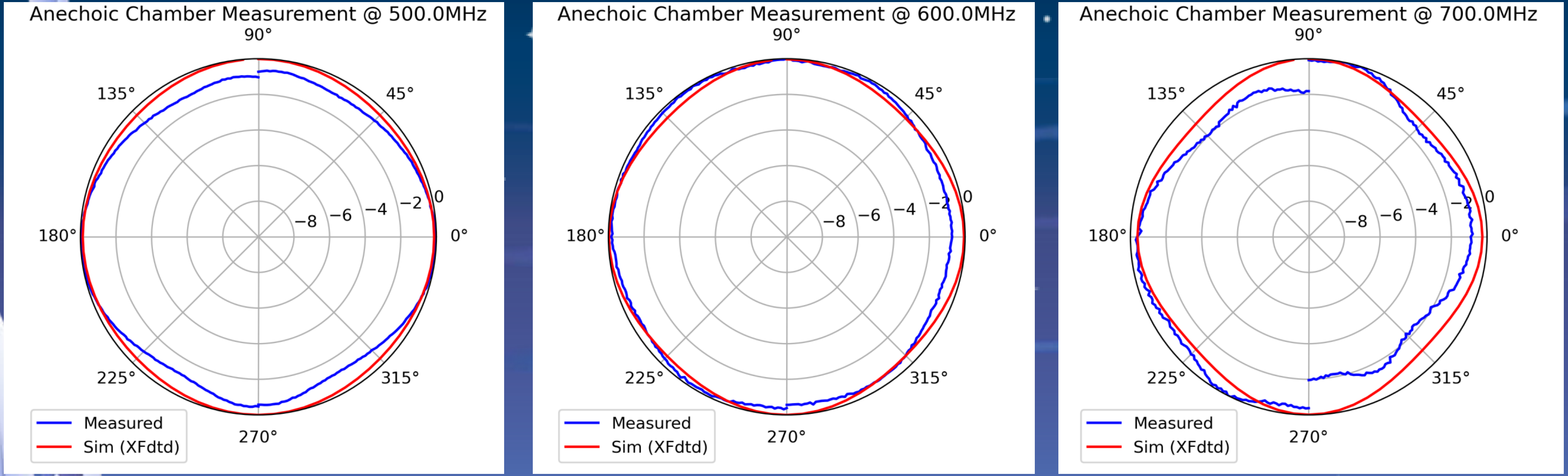
Industry standard for S11 is derived from a **target of 2-3 VSWR**:

- 2 VSWR is -9.55 dB S11
- 3 VSWR is -6 dB S11

Equation for S11:

$$\Gamma = \frac{Z_{in} - Z_o}{Z_{in} + Z_o}$$

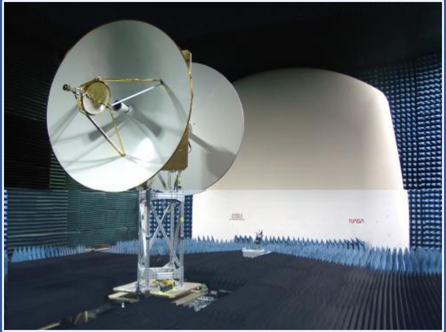
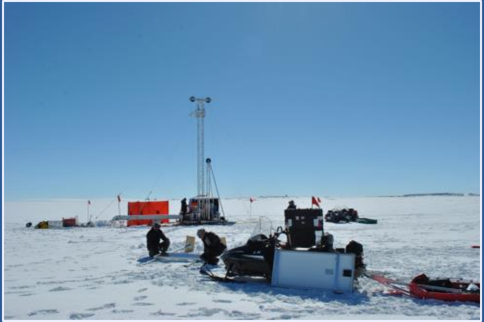
Gain vs Frequency (in-air)



- Azimuthal plane
- Azimuthally symmetric in gain
- The closer a line is to the outer edge (0 dB), the stronger the signal
- Frequency dependent

In-ice plots?

How antenna data is gathered….

	In-air	In-ice
Simulation	XFdtd®	XFdtd®
Measured	 	 ?

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Calibration Efforts

Antenna Calibration

- Want to characterize antenna performance with calibration, not just rely on simulation
- One calibration technique: measure VEL

What does VEL/effective height tell us:

- Calculate E^i of neutrino
- How much energy the source has
- Antenna gain/antenna pattern

Vector Effective Length (VEL)

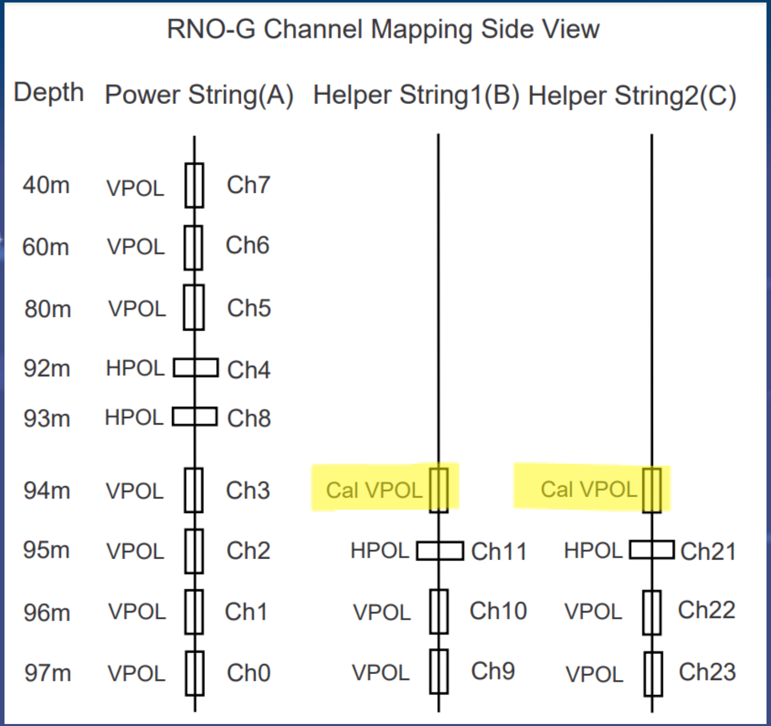
Also called 'effective height'. This is the antenna response as a function of frequencies.

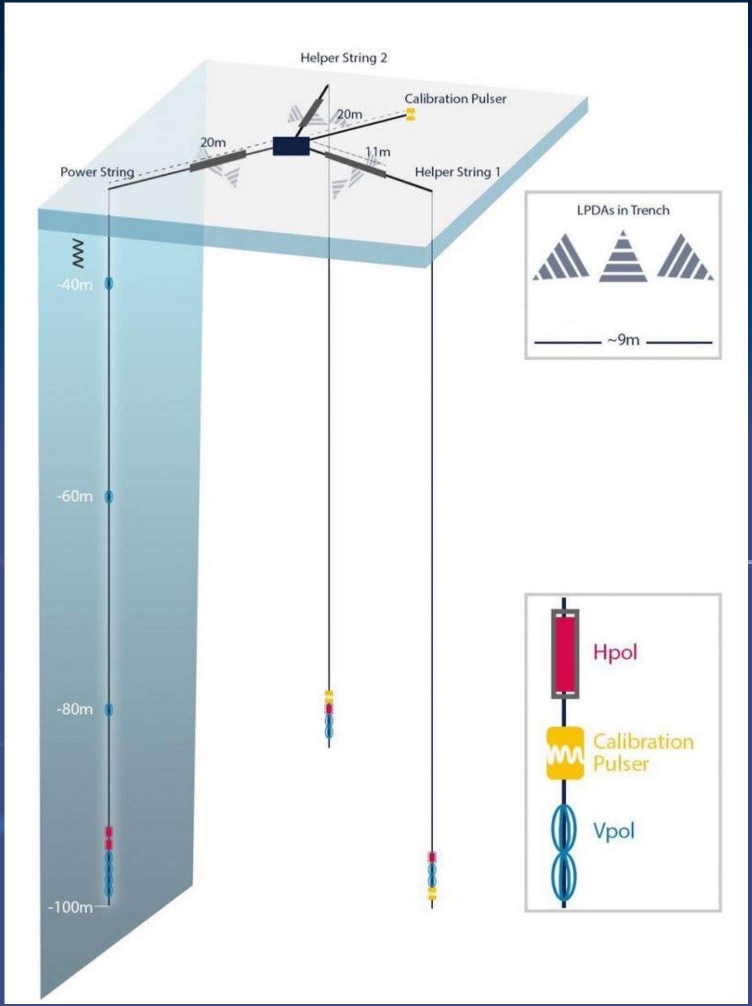
Equation: $\mathrm{VEL} = \frac{V^{oc}}{E^i}$

- V^{oc} - voltage read on antenna terminal in frequency space (after Fourier transform)
- E^i - incident electric field, has polarization (the incident angle)
- VEL is a function of frequencies AND polarization.

Calibration Pulser

- Cal pulser in ice that only pulses in vpol
- Certain information we can gain using cal pulser





How we get VEL

Simulated Data

Uses math and assumptions

↓

Simulated VEL

**Both in ice and in air simulations have been done

Physical Data (this is hard)

In-air (anechoic chamber)

Send pulser

Measure V^{oc} and E^{i}

↓

Calculate VEL

In-ice

So far we have only used simulated in-ice VEL

Measuring VEL is **very difficult.**

- Borehole size constraints
- There are only vpol cal pulsers
- Cal pulser is fixed in place

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Potential Pipeline

In-ice

Send Cal Pulser

Measure V^{oc}_{cp} and get E^{i}_{cp} with simulated VEL

Account for E^{i}_{cp} propagates in ice through simulation to get E^{i}_{vpol}

Measure V^{oc}_{vpol}

Calculate VEL

Bigger Picture for VEL:

Understand antenna performance and calculate E^{i} for neutrino

- Kind of goes in circles
- The equation is re-used many times
- Used with simulated VEL and cal pulser

A reminder, the equation is: $\mathrm{VEL} = \frac{\mathrm{V}^{\mathrm{oc}}}{\mathrm{E}^{\mathrm{i}}}$

1 **Measured VEL**

Specific to the antenna, needed to calculate other parameters in equation

2 **Read Voltage**

Peak in voltage data when neutrino interacted with antenna

3 **Use VEL, V get E**

Calculate incident electric field of neutrino signal

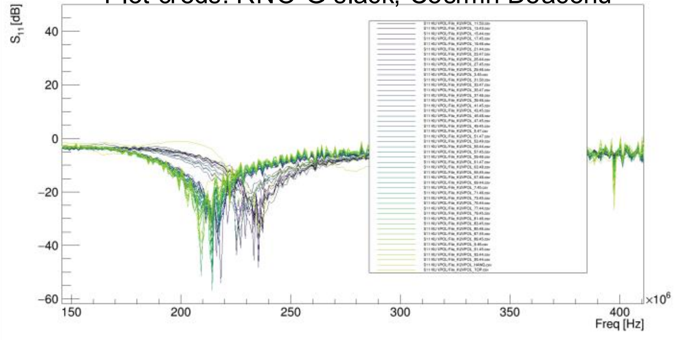
4 **Determine direction**

Triangulate and find the neutrino direction

Vertically Polarized (Vpol)

Horizontally Polarized (Hpol)

Plot creds: RNO-G slack, Cosmin Deaconu



Vpol

What has been done:

- S11 data gathered 2024 cycle with a VNA in a separate borehole with IDL pulser
- Not much details

What could be done?

- Utilize vpol cal pulser and simulated VEL to find cal pulsar E^{i}_{cp}. Find E^{i} through raytracing and ice attenuation. Calculate actual VEL (red pipeline I showed)

Hpol

What has been done:

- S11 data gathered 2024 cycle with a VNA in a separate borehole with IDL pulser
- In 2022 vpol/hpol measurements from FID pulser

What could be done?

- Tricker problem as we only have a vpol calibration pulser
- Hpol antennas are not perfect, so it could pick up vpol signal: "cross pol"
- Same analysis may be possible to do with cross pol data

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Open Ended Questions:

1. Can VNA be used to measure VEL?

2. There is S11 data (KU Team) from 2024 installment for both vpol and hpol, any idea how this might’ve been done or does anyone know anymore details?

3. **Why is there no hpol cal pulser?** Cost reasons maybe? No vertical design for an hpol cal pulser?

Resources

IMAGES

- https://radio.uchicago.edu/wiki/index.php/Science_Antennas#Downhole_H-Pol_Antenna
- https://en.wikipedia.org/wiki/Radio_Neutrino_Observatory_Greenland#/media/File:RNO-G_station_sketch.jpg
- https://www.bitweenie.com/listings/the-decibel
- https://www.appec.org/news/deployment-of-the-radio-neutrino-observatory-greenland-rno-g-started

PAPERS

- https://arxiv.org/pdf/2411.12922
- RNO-G HPol Design
- https://arxiv.org/pdf/2010.12279
- https://digitalcommons.calpoly.edu/theses/2888/

PEOPLE (I asked questions from): Everett, Zoe, Melanie, Alisa

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Thanks!

Questions?

Γ	Γ (dB)	VSWR	Mismatch Efficiency
0	-Infinity	1.0	1
0.1	-20	1.22	0.99
0.25	-12	1.67	0.94
0.33	-10	2.0	0.89
0.5	-6	3.0	0.75
0.6	-4.4	4.0	0.64
0.67	-3.5	5.0	0.55
0.9	-0.9	19.0	0.19